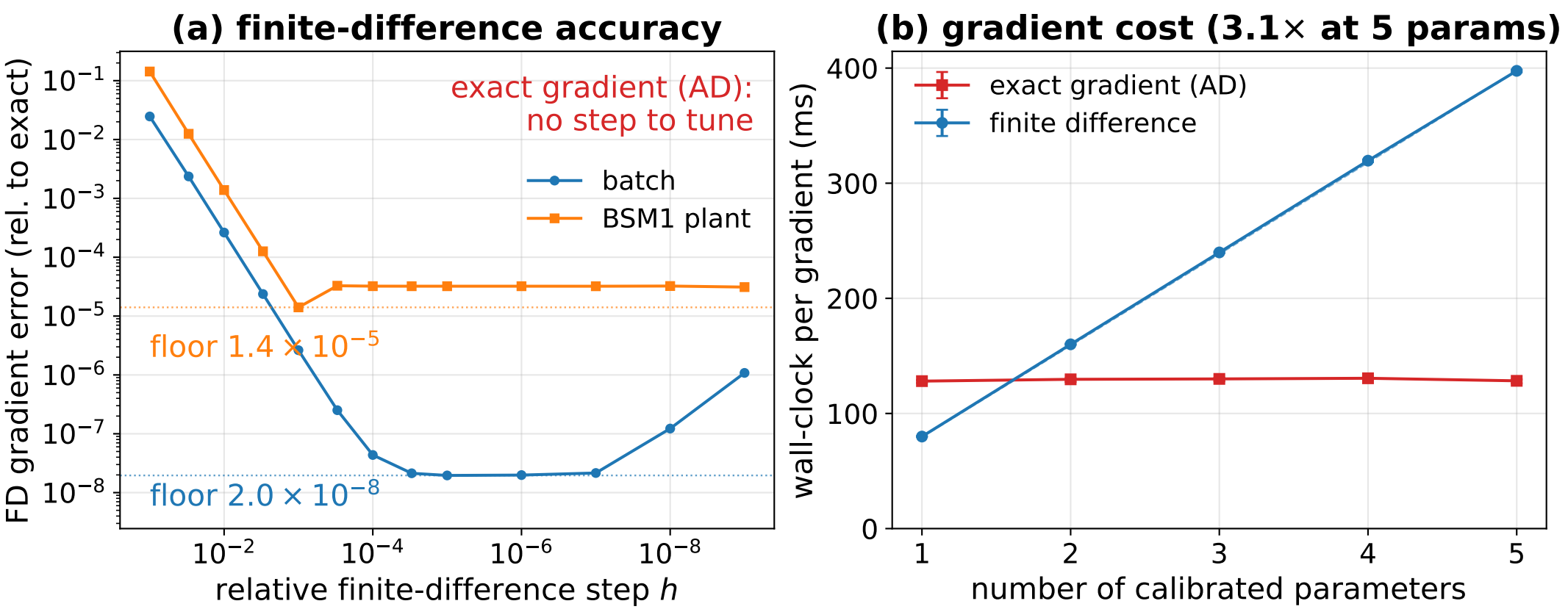
(a) finite-difference accuracy
FD gradient error (rel. to exact)
relative finite-difference step h
exact gradient (AD):
no step to tune
batch
BSM1 plant
floor 1.4 × 10⁻⁵
floor 2.0 × 10⁻⁸
(b) gradient cost (3.1× at 5 params)
wall-clock per gradient (ms)
number of calibrated parameters
exact gradient (AD)
finite difference